The Hole Rejects the Fit

A Gauge-Invariant Health-Bar for Dangerous Estimation

Matěj Rada
MatejRada@email.cz

July 21, 2026

Abstract

Resolution Geometry (RG) splits an observation system into a *resolved* sector, which the data constrain, and a *null* sector—the "hole"—which they do not. This note asks whether that hole can be used to *reject dangerous fits*: estimates that look good by ordinary criteria but are actually determined by directions the data leave silent. The answer is yes, and the honest accounting is stated first. The machinery reduces entirely to established objects—the resolved/null split is the model-resolution matrix of inverse theory (Backus–Gilbert), its noise-aware form is Fisher/Cramér–Rao, the coupling floor is the Schur complement, and the similarity invariants are canonical correlations. RG invents no new detector. What it contributes is a single *coherent, gauge-invariant, per-target* health-bar that *no single standard diagnostic reproduces*, together with a reject-by-default discipline: report the resolved part, quarantine the hole, and score the target's safety margin. We demonstrate this on a battery where each ordinary diagnostic, used alone, *accepts* a fit the health-bar rejects. A fit with global $R^2 = 0.95$ carries a target whose health-bar is 0.00 (the target lives in the hole)—R^2 misses it. A map with condition number exactly 1.00 carries a target the noise makes unidentifiable, health 0.00—the condition number misses it. And when transporting a calibration to a new domain, raw marginal similarity picks the wrong source (transport error 3.2) while RG's gauge-invariant coupling alignment picks the safe one (error 0.37). The hole is not a nuisance to be regularized away; treated as geometry, it is the instrument that refuses the fits most likely to hurt you.

1 The question, and the honest reduction

A "dangerous fit" is one that reports a confident estimate of something the data do not determine: a quantity living in the null sector (the hole), dressed up as resolved by a good global fit, a benign-looking condition number, or a prior that a regulariser has quietly supplied. The question is whether RG's hole geometry gives a principled way to *detect and refuse* such fits, and how that differs from standard practice.

Remark 1 (What this is *not* — stated up front). Every object used below is established. The resolved/null decomposition is the *model-resolution matrix* $\mathbf{R} = J^{+}J$ of geophysical inverse theory (Backus–Gilbert); its noise-aware form is the *Fisher information* $F = J^{\top}\Sigma^{-1}J$ and the Cramér–Rao bound; the coupling floor is the *Schur complement*; the similarity invariants are *canonical correlations*. RG contributes no new primitive here, exactly as the elimination result requires.

Remark 2 (What is nonetheless different). Three things, none of which is a single standard diagnostic:

1. **Reject-by-default.** Ordinary estimators *fill* the hole—least squares with unbounded variance, ridge/Bayes with a prior that can look confident. RG's default output reports the resolved projection, *quarantines* the null component, and scores the target.

2. **A per-target, noise-aware health-bar.** A single bounded scalar $H(c) \in [0, 1]$ for the specific quantity $g = c^\top\theta$ you care about—not a global fit metric, not a scalar property of J alone.

3. **Gauge-invariance.** Because H and the resolved directions are coordinate-free, they decide *similarity and transport safety* where raw statistics mislead.

The claim is coherence, not novelty: one geometry yields all three consistently, and each ordinary diagnostic, alone, misses at least one danger the health-bar catches.

2 The health-bar and the rejection protocol

Definition 1 (Fit health-bar)**.** For forward map J, noise covariance Σ, and target functional $g = c^\top\theta$, form the Fisher information $F = J^\top\Sigma^{-1}J = \sum_i \lambda_i u_i u_i^\top$. Fix an information floor τ (the prior/data crossover; $\tau = 1$ in prior-variance units). With $\hat{c} = c/\|c\|$,

$$H(c) = \sum_{i:\, \lambda_i \geq \tau} (u_i^\top \hat{c})^2 \ \in [0, 1],$$

the fraction of the target that lies in the resolved sector (the resolution matrix's diagonal in direction c, noise-aware). $H = 1$: fully resolved. $H \to 0$: the target lives in the hole.

Definition 2 (Rejection protocol)**.** Report $\hat{g}$ only through the resolved projection of c; quarantine the null component and do *not* report it as determined. Publish $H(c)$ as the safety margin and the resolved-sector variance (with the Schur floor as the irreducible remainder). Reject—refuse to estimate—any target with $H(c)$ below a chosen margin.

3 The battery: each ordinary diagnostic accepts what the health-bar rejects

Finding 1 (R^2 misses an unresolved target)**.** A linear model with singular values $(10, 6, 3, 0.02)$ and moderate noise fits well globally—$R^2 = 0.95$, the ordinary "trust it" verdict. Yet the target aligned with the weak direction has health-bar $H = 0.00$ (hole-exposure 1.00), while a strong-direction target has $H = 1.00$. The same fit is safe for one target and dangerous for another; a global fit metric cannot see the difference, the per-target health-bar does.

Finding 2 (The condition number misses a noise-killed target)**.** A map with orthonormal columns has $\mathrm{cond}(J) = 1.00$—"well-posed" by the standard scalar. Place large measurement noise exactly along the observation direction that reads out the target c. Then $\mathrm{cond}(J)$ is unchanged, but the noise-aware health-bar is $H = 0.00$: the target is unidentifiable. Condition number looks at J alone; the health-bar looks at J and Σ together, and refuses.

Finding 3 (Similarity: raw statistics pick the wrong source, geometry picks the safe one)**.** A calibration is to be transported to a new domain T. Two candidate sources: R_1 (couples to feature 1) and R_2 (couples to feature 4). T couples like R_1 but has marginal statistics like R_2.

	transport error to T	naive marginal-distance	RG coupling-alignment
source R_1 (safe)	0.37	9.64 (far)	0.99 (aligned)
source R_2 (unsafe)	3.22	4.29 (near)	0.08 (orthogonal)

Raw marginal similarity picks R_2 and transports an estimate that is nearly ten times worse; RG's gauge-invariant coupling alignment picks R_1, the source that actually transports. The "hole geometry" orients correctly in similarity because it compares what is resolved, not what is superficially alike.

4 Why this is the right last battery

The earlier necessity work showed *when* the coherent geometry is needed; this shows *what it protects you from*. A fit is dangerous exactly when its apparent quality is carried by the hole—by directions the data do not constrain, by noise the map ignores, or by a resemblance that is not geometric. Each of those three dangers is invisible to one of the field's standard scalars (R^2, cond, marginal similarity) and visible to the coherent, gauge-invariant health-bar. RG's contribution is not a new inequality but a *stance*: make the hole primary, refuse to estimate through it, and score every target and every transport by how much of it is genuinely resolved. That stance is what turns a pile of established identifiability facts into a safety instrument.

Non-claims

Non-claim 1. No new mathematical primitive. Resolution matrix, Fisher/Cramér–Rao, Schur complement, and canonical correlations are all classical; the health-bar is the resolution-matrix diagonal made noise-aware, per-target, and gauge-invariant.

Non-claim 2. The demonstrations are controlled synthetic constructions chosen so that each ordinary diagnostic provably fails; they show the health-bar catches those cases, not that real fits always contain them.

Non-claim 3. No physical claim. This is estimation safety—identifiability, resolution, transport—not a statement about any domain's mechanisms.

References

[1] G. Backus and F. Gilbert, "The resolving power of gross Earth data," *Geophys. J. R. Astron. Soc.* 16 (1968), 169–205.

[2] C. R. Rao, "Information and accuracy attainable in the estimation of statistical parameters," *Bull. Calcutta Math. Soc.* 37 (1945), 81–91.

[3] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.

[4] H. Hotelling, "Relations between two sets of variates," *Biometrika* 28 (1936), 321–377.